



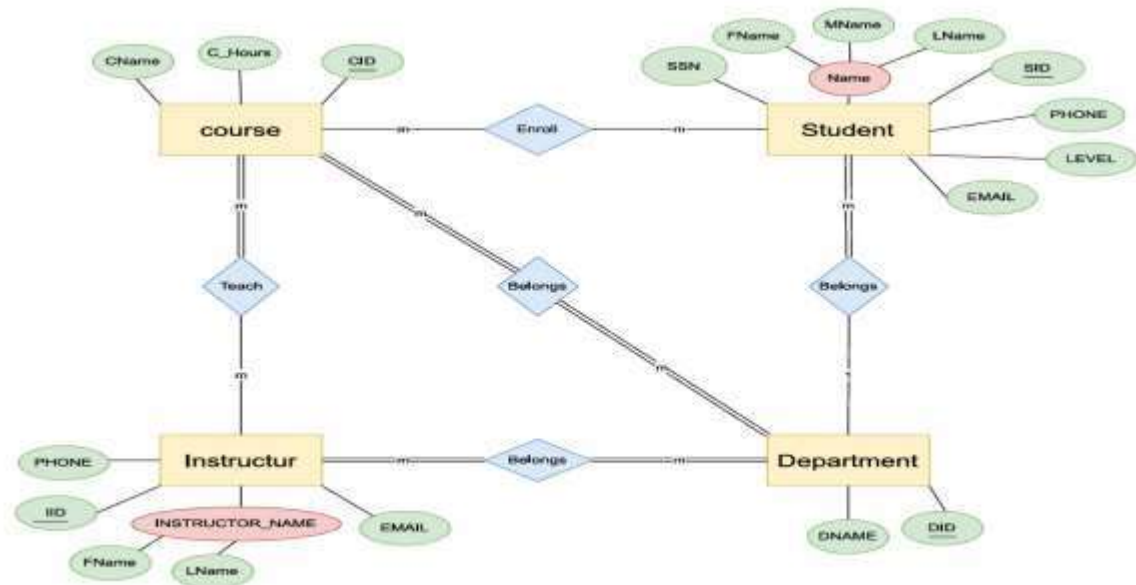
Report DB project

University Course Registration

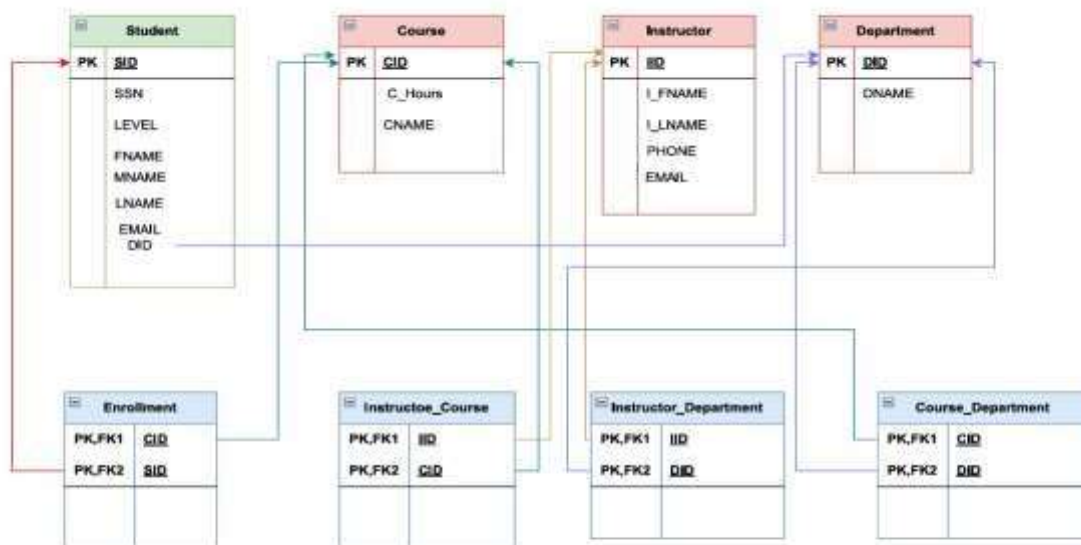


THE TEAM
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First we made the ERD



Then we made the Relational Schema from the ERD



Then we implemented the sql code and every one of us made part of it (DDL,DML,DQL) .

```
----- DDL_CODE-----
create database University_Course_Registration ;
use University_Course_Registration;
create table department (
DID INT PRIMARY KEY,
DName VARCHAR(50) NOT NULL UNIQUE
);
create table student (
SID int primary key AUTO_INCREMENT,
SSN int unique ,
level int,
FName VARCHAR(50) NOT NULL ,
MName VARCHAR(50) NOT NULL ,
LName VARCHAR(50) NOT NULL,
phone varchar(11),
Email varchar(200),
DID INT,
FOREIGN KEY (DID) REFERENCES department(DID) ON DELETE SET NULL,
CONSTRAINT chk_student_phone CHECK (phone REGEXP '^01[0-9]{9}$'),
CONSTRAINT chk_level CHECK (level BETWEEN 1 AND 4),
CONSTRAINT chk_student_email CHECK (Email REGEXP '^([A-Za-z0-9._%-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,})$')
);
create table course (
CID INT PRIMARY KEY,
CName VARCHAR(50),
C_Hours INT CHECK (C_Hours IN (0,2,3))
);
create table instructor(
IID int primary key,
FName VARCHAR(30) NOT NULL,
LName VARCHAR(30) NOT NULL,
phone varchar(11),
Email varchar(200),
CONSTRAINT chk_instructor_phone CHECK (phone REGEXP '^01[0-9]{9}$'),
CONSTRAINT chk_instructor_email CHECK (Email REGEXP '^([A-Za-z0-9._%-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,})$')
);
```

```

CREATE table enrollment(
SID INT,
FOREIGN KEY(SID) REFERENCES student(SID) ON DELETE CASCADE ON UPDATE CASCADE,
CID INT,
FOREIGN KEY( CID) REFERENCES course( CID) ON DELETE CASCADE ON UPDATE CASCADE,
primary key(SID,CID)
) ;

CREATE table instructor_course(
IID int ,
CID int,
primary key (IID,CID),
foreign key(CID) references course(CID) ON DELETE CASCADE ON UPDATE CASCADE,
foreign key(IID) references instructor(IID) ON DELETE CASCADE ON UPDATE CASCADE
) ;

CREATE table instructor_dep(
IID int ,
DID int,
primary key (IID,DID),
foreign key(IID) references instructor(IID) ON DELETE CASCADE ON UPDATE CASCADE,
foreign key(DID) references department(DID) ON DELETE CASCADE ON UPDATE CASCADE
) ;

CREATE table course_department(
DID int ,
CID int,
primary key (CID,DID),
foreign key(CID) references course(CID) ON DELETE CASCADE ON UPDATE CASCADE,
foreign key(DID) references department(DID) ON DELETE CASCADE ON UPDATE CASCADE
) ;

alter table student modify SSN varchar(14);
alter table student add constraint check_SSN_LENGTH check(length(SSN)=14);

```

----- DML_CODE-----

-- 1

INSERT INTO department (DID, DName) VALUES

(1, 'Computer Science'),
(2, 'Information Technology'),
(3, 'Software Engineering'),
(4, 'Data Science'),
(5, 'Cyber Security'),
(6, 'Artificial Intelligence'),
(7, 'Computer Engineering'),
(8, 'Information Systems'),
(9, 'Network Engineering'),
(10, 'Web Development'),
(11, 'Mobile Computing'),
(12, 'Cloud Computing'),
(13, 'Database Systems'),
(14, 'Game Development'),
(15, 'Computer Graphics'),
(16, 'Human Computer Interaction'),
(17, 'Bioinformatics'),
(18, 'Computer Networks'),
(19, 'Digital Marketing'),
(20, 'E-commerce'),
(21, 'Business Informatics'),
(22, 'Health Informatics'),
(23, 'Robotics'),
(24, 'Internet of Things'),
(25, 'Quantum Computing'),
(26, 'Computer Architecture'),
(27, 'Operating Systems'),
(28, 'Compiler Design'),

----- DQL_CODE-----

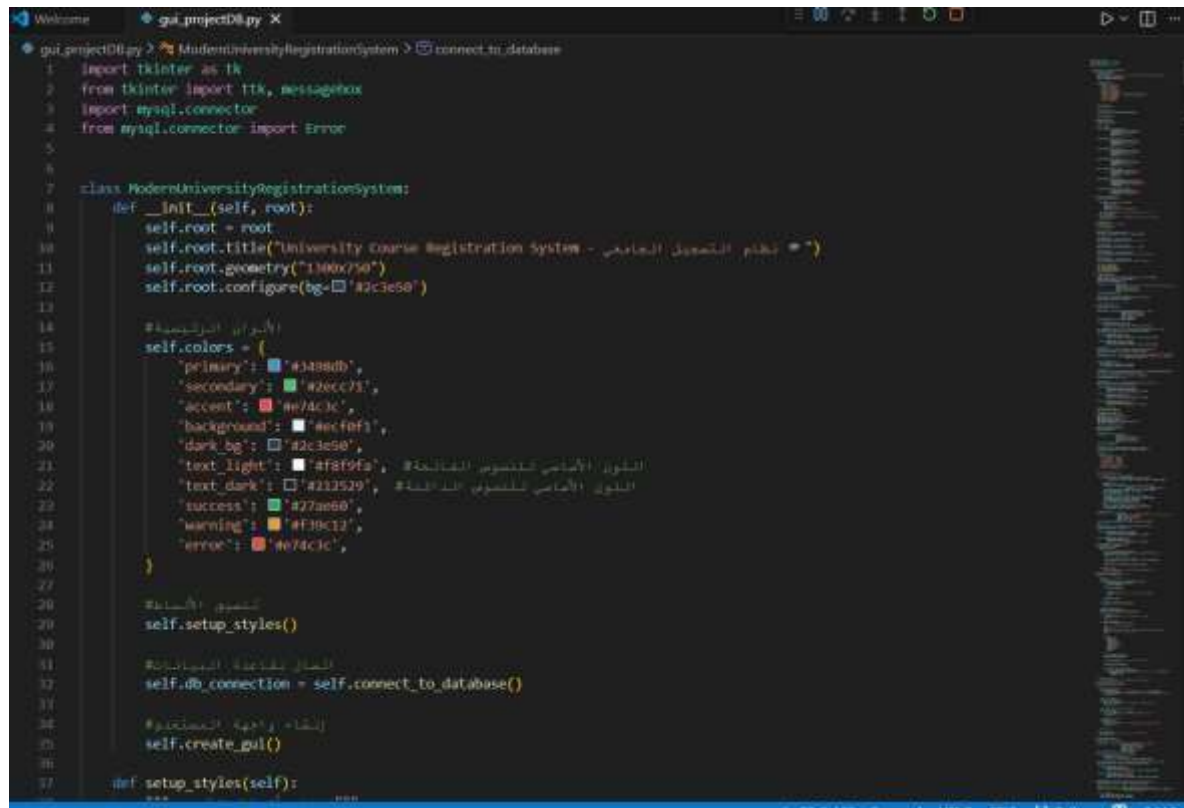
```
select * from student;
select * from course;
select * from course_department;
select * from enrollment;
select * from instructor;
select * from instructor_dep;
select * from instructor_course;
select * from department;
```

```
SELECT S.FName, S.LName, D.DName
FROM student S
JOIN department D ON S.DID = D.DID;
```

```
SELECT SID, FName, LName, DID
FROM student;
```

```
SELECT IC.IID, IC.FName, IC.LName, C.CName
FROM instructor IC
join instructor_course I_C ON IC.IID=I_C.IID
JOIN course C ON C.CID=I_C.CID
;
select * from department;
```

After we had made the database, we started in GUI using python.



```
gui_projectDB.py X
gui_projectDB.py > MudentUniversityRegistrationSystem > connect_to_database
1 import tkinter as tk
2 from tkinter import ttk, messagebox
3 import mysql.connector
4 from mysql.connector import Error
5
6
7 class MudentUniversityRegistrationSystem:
8     def __init__(self, root):
9         self.root = root
10        self.root.title("University course registration system - نظام التسجيل الجامعي")
11        self.root.geometry("1300x750")
12        self.root.configure(bg="#2c3e50")
13
14        # الألوان الرئيسية
15        self.colors = {
16            'primary': "#4388db",
17            'secondary': "#2ecc71",
18            'accent': "#e74c3c",
19            'background': "#ecf0f1",
20            'dark_bg': "#2c3e50",
21            'text_light': "#f8f9fa", # اللون الأبيض للتصميم الفاتح
22            'text_dark': "#212529", # اللون الأسود للتصميم الداكن
23            'success': "#27ae60",
24            'warning': "#f39c12",
25            'error': "#e74c3c",
26        }
27
28        # التصميم
29        self.setup_styles()
30
31        # الاتصال بقاعدة البيانات
32        self.db_connection = self.connect_to_database()
33
34        # إنشاء واجهة المستخدم
35        self.create_gui()
36
37    def setup_styles(self):
38        ...
39        ...
```


After all this effort this is the final output

University Course Registration System

التسجيل في المقررات | إدارة المقررات | إدارة الطلاب

بحث الطلاب

بحث بالاسم

بحث

عرض الكل

SID	SSN	PHName	MName	LName	Level	Department	Phone	Email
1	023-45-6789-01	Ahmed	Muhammad	Ali	1	Computer Science	01012345678	ahmed.ali@university.edu
2	234-56-7890-12	Muhammad	Muhammad	Hassan	2	Information Technology	01023456789	mohamed.hassan@university.edu
3	345-67-8901-23	Muhammad	Omar	Walid	3	Software Engineering	01034567890	mohamed.walid@university.edu
4	456-78-9012-34	Omar	Muhammad	Sayed	4	Data Science	01045678901	omar.sayed@university.edu
5	567-89-0123-45	Muhammad	Ahmed	Tarek	1	Cyber Security	01056789012	mohamed.tarek@university.edu
6	678-90-1234-56	Ahmed	Walid	Hassan	2	Artificial Intelligence	01067890123	ahmed.hassan@university.edu
7	789-01-2345-67	Walid	Hassan	Kamel	3	Computer Engineering	01078901234	walid.kamel@university.edu
8	890-12-3456-78	Hassan	Tarek	Rayad	4	Information Systems	01089012345	hassan.rayad@university.edu
9	901-23-4567-89	Tarek	Hassan	Sami	1	Network Engineering	01090123456	tarek.sami@university.edu
10	012-34-5678-90	Youssef	Hassan	Adel	2	Web Development	01001234567	youssef.adel@university.edu
11	112-33-3333-44	Hassan	Kareem	Fahd	3	Mobile Computing	01011233333	hassan.fahd@university.edu
12	223-33-4444-55	Kareem	Shadi	Gerasi	4	Cloud Computing	01022333333	kareem.gerasi@university.edu
13	333-44-5555-66	Shadi	Hadi	Hady	1	Database Systems	01033344445	shadi.hady@university.edu
14	444-55-6666-77	Hadi	Rayad	Issam	2	Game Development	01044455556	hadi.issam@university.edu
15	555-66-7777-88	Rayad	Sami	Adel	3	Computer Graphics	01055566667	rayad.adel@university.edu
16	666-77-8888-99	Sami	Muhammad	Sami	4	Human Computer Interaction	01066677778	sami.muhammad@university.edu
17	777-88-9999-00	Muhammad	Adel	Leif	1	Biometrics	01077788889	mohamed.leif@university.edu
18	888-99-0000-11	Adel	Tarek	Muhammad	2	Computer Networks	01088899990	adel.muhammad@university.edu

إضافة طالب جديد

تعديل بيانات طالب

حذف طالب

University Course Registration System

التسجيل في المقررات | إدارة المقررات | إدارة الطلاب

التسجيل في المقررات

رقم طالب

التسجيل في المقررات

الطلاب الذين قاموا بالتسجيل في المقررات

المقررات المتاحة

المقررات المسجلة

المقررات المسجلة

المقررات المتاحة

التسجيل في المقررات

إلغاء التسجيل

التسجيل في المقررات

University Course Registration System - نظام التسجيل الجامعي

المقايير والمصطلحات | التسجيل في المقررات | إدارة المقررات | إدارة الطلاب

التقارير المتاحة

عرض جميع المقررات	عرض جميع المقررات
الطلاب حسب القسم	المقررات حسب القسم

قائمة جميع المقررات

- العدد: 195
رقم المقرر: 195
اسم المقرر: Computer Networks
المستوى: 3
المقرر: Artificial Intelligence
المدرس: Dr. Amr Shamsy
- العدد: 111
رقم المقرر: 111
اسم المقرر: Human Computer Interaction
المستوى: 3
المقرر: Bioinformatics
المدرس: Dr. Mostafa Kamel
- العدد: 171
رقم المقرر: 171
اسم المقرر: Internet of Things
المستوى: 3
المقرر: Business Informatics
المدرس: Dr. Zaki Osman

العدد: 48

جميع التقارير

تقرير المقرر

طباعة التقرير